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FURNACE-X1 v1.0.0: AI Weight-Streaming and Secure Storage SoC

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Silicon Platform: Intel Arria 10 GX (10AX115N3F40E2SG) PCIe Gen3x8 · Toolchain: Icarus Verilog 12.0 / Yosys 0.33 / Quartus Prime 17.1 STA · 30 October 2026 (2026-10-30 02:55 UTC+8 Beijing Time)

Abstract

FURNACE-X1 integrates standard LZ4 frame decompression, quarantined AES-128/GCM authenticated decryption, an eight-lane W4A16/BF16 tensor engine, AXI4-Lite control, a hardware command ring and a four-initiator/four-target AXI4 crossbar. Software encrypts weights before framing the ciphertext with LZ4; hardware decompresses before decrypting and computing. Software-vs-Hardware numerical comparison was bit-exact for the simulated RTL: nine integrated configuration/seed runs checked 216 descriptors, comprising 180 successful tiles and 36 authentication rejections. Separate tensor testing transferred 256 vectors per mode. L1 RTL timing lint passed; L2 Intel Quartus Prime 17.1 TimeQuest post-fit STA achieved WNS_setup = +1.464 ns at a 10.0 ns clock period. L3 physical Arria 10 FPGA HIL recorded three deterministic passing runs. Its retained observations are diagnostic register reads, not a tile-level physical numerical-comparison ledger [E048–E056].

Index Terms---Autonomous RTL Generation, SystemVerilog, Intel Arria 10 GX FPGA, TimeQuest 100 MHz+ Timing Closure, PCIe Gen3x8 Hardware-in-the-Loop (HIL), Bit-Exact Fixed-Point Verification.

I Architecture and design rationale

The top-level furnace_x1_top uses one synchronous clock domain and one descriptor executor. Serial tile execution simplifies authenticated-publication ownership and accumulator isolation while retaining independently arbitrated crossbar traffic. Four DMA instances serve compressed input, authenticated writeback, tensor operands/results and descriptors/completions. Single-beat DMA transactions favor bounded ownership over peak bandwidth. Registered history reads, split AES rounds, sequential GHASH and staged floating-point arithmetic bound combinational depth. The following blocks are arranged row-major in three columns; arrows distinguish control, memory transactions and provisional or authenticated data [E001–E016].

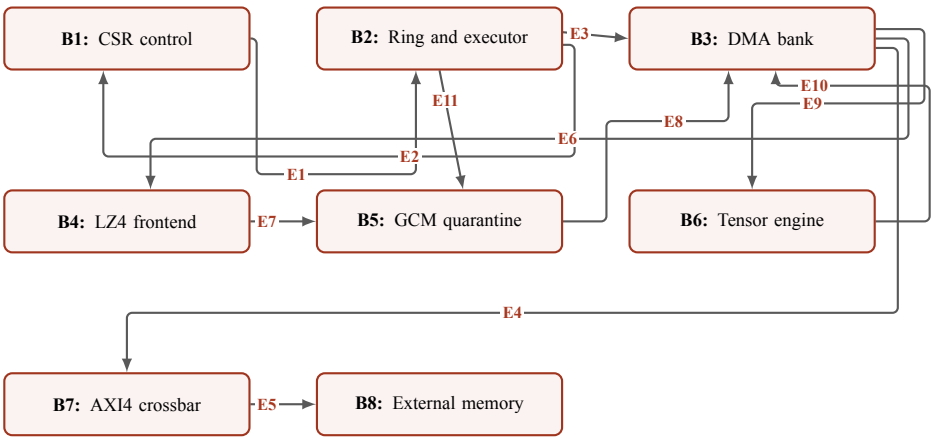


Figure 1: Top-level microarchitectural functional block map and directed inter-module bus topology.

Block	Responsibility
B1: CSR control	AXI4-Lite registers manage enable, committed key, ring base, doorbell, status and sticky interrupts.
B2: Ring and executor	Tracks producer/consumer counters, validates descriptors and sequences one tile through completion or terminal fault.
B3: DMA bank	Four independent initiators translate byte streams into aligned 128-bit memory transactions.
B4: LZ4 frontend	Parses frames, verifies XXH32 checksums and reconstructs ciphertext using fixed 65536-byte circular history.
B5: GCM quarantine	Pipelined AES-128 CTR and sequential GHASH retain plaintext privately until tag and complete-frame validation succeed.
B6: Tensor engine	Eight integer or BF16 lanes feed registered balanced reductions and a dependency-controlled accumulator.
B7: AXI4 crossbar	Independent per-target read/write round-robin arbitration retains ownership through responses and routes IDs.
B8: External memory	Four target-facing ports hold descriptors, framed ciphertext, plaintext, activations, results and completions.

Directed relationship	Meaning
E1: B1 to B2	Configuration and doorbell
E2: B2 to B1	Completion status and IRQ events
E3: B2 to B3	Validated DMA requests
E4: B3 to B7	Four AXI initiators
E5: B7 to B8	Four bidirectional AXI target links
E6: B3 to B4	Framed ciphertext stream
E7: B4 to B5	Provisional ciphertext and frame verdict
E8: B5 to B3	Authenticated plaintext writeback
E9: B3 to B6	Weight and activation vectors
E10: B6 to B3	Final tensor result
E11: B2 to B5	Key, nonce, tag and AAD

II Interfaces and configuration

Signal / group	Direction	Width	Contract
clk, rst_n, irq	Input clock/ reset; output interrupt	1 bit each	100 MHz baseline; active-low asynchronous reset with synchronous release assumed. Reset zeros outward controls and payloads, cancels ownership and requires connected AXI endpoints to reset concurrently. IRQ combines enabled sticky success/ failure events.
s_axil_*	AXI4-Lite slave	16-bit address; 32-bit data; 4-bit WSTRB	Complete AW/W/B/AR/R channels, including 3-bit protection and 2-bit responses. AW and W are decoupled; one write and one read may be pending concurrently. Invalid, unaligned, read-only or forbidden configuration writes return SLVERR.

Signal / group	Direction	Width	Contract
m_axi_*	Four AXI4 masters	Packed [3:0] ports; 32-bit address; 128-bit data; 4-bit ID	Complete channels include LEN8, SIZE3, BURST2, LOCK1, CACHE4, PROT3, QOS4 and REGION4, without USER. Address[31:30] selects a target; the full address is forwarded. ID concatenates the two-bit initiator index and two-bit local ID.
Internal byte and block interfaces	Producer/ consumer	8-bit data/ error; 32-bit lengths	Streams use valid/ready/data/last in increasing address order. Block requests use start_valid/start_ready and held done_valid/error. Stalled payloads remain stable. The GCM frame_done interface separately fences successful LZ4 completion.
Crypto and tensor leaf ports	Request inputs; result outputs	Key/tag128; nonce96; AAD416; tensor operands128; result64	Crypto numeric ports use conventional most-significant-byte-first notation; AAD byte i occupies bits 8*i+:8. Tensor lane i uses activation bits 16*i+:16; BF16 weights use corresponding fields and W4 weights use bits 4*i+:4.

Parameter	Default	Meaning / supported values
MAX_TILE_BYTES	4096	Integer 1..65536; bounds integrated decoded ciphertext and plaintext quarantine. It does not include LZ4 framing overhead. Small legal settings reject non-fitting tensor descriptors.
RING_DEPTH	8	Integer 1..16, including non-powers of two. Physical slots wrap with explicit counters.
furnace_lz4_ frontend.MAX_ DECODED_BYTES	65536	Standalone integer 1..131072; total frame-output bound with unchanged 65536-byte history. The integrated top explicitly binds it to MAX_TILE_BYTES. Illegal parameters produce PARAMETER ERROR diagnostics.

III Behavior and corner cases

A Frame decoding and publication fence

Accepts standard magic 04 22 4d 18, version-01 FLG and BD=0x40, raw/compressed blocks, linked/ independent history, optional content size and block/content checksums. Repeated-255 length extensions and overlapping offsets 1..65535 are supported. Dictionary IDs, unsupported flags, malformed terminal sequences, trailing/concatenated frames and inconsistent lengths/last are rejected. Leaf bytes remain provisional until successful completion; a terminal error preserves an already stalled output beat before completing. Empty leaf output is legal, but integrated tensor tiles are nonempty.

B Authenticated decryption

AES-128 uses genuine key expansion and ten rounds. For the 96-bit nonce, J0=nonce||1 and payload counters begin at two. GHASH authenticates fixed 52-byte descriptor AAD, separately padded ciphertext and big-endian bit lengths. All 128 tag bits are compared. Plaintext publication requires exact length, successful frame validation and authentication; tag failure permits no plaintext or result writes. AES block output becomes valid post-E21, key expansion takes 40 cycles and GHASH multiplication takes 128 recurrence cycles.

C Numerical contract

Mode 0 multiplies signed W4 (-8..7) by signed A16 (-32768..32767), reduces exact products and accumulates modulo 2^64. Mode 1 uses BF16 weights/activations; mode 2 uses signed W4 weights and BF16 activations. Each floating-point product, balanced-tree addition and accumulator addition rounds binary32 nearest ties-to-even. Gradual underflow, signed zero, infinities and overflow are supported; NaN/invalid results canonicalize to 0x7fc00000. Floating results occupy low32, with high32 zero. Mode 3 is illegal.

D Tensor timing

E0 accepts a vector; multiplication occupies E1..E4, three reduction levels E5..E19 and accumulator addition E20..E24. Integer mode follows the same observable schedule. Only one vector remains uncommitted at a time. Last publishes post-E24; earliest retirement is E25. Intermediate vectors do not publish results, and final results remain stable under backpressure.

E Interconnect and fault containment

Legal AXI transactions are 16-byte aligned, SIZE=4 INCR bursts of 1..16 beats without 4-KiB or target-region crossing. Invalid reads return DECERR beats; invalid writes drain without forwarding. Ownership persists through RLAST or B, and continuously eligible contenders receive service within four completed arbitration turns. DMA uses single-beat transfers and exact final write strobes. Downstream errors become tile faults; already accepted authenticated writes cannot be rolled back.

F Command and completion ABI

Each 128-byte little-endian descriptor contains opcode at0, flags4, frame address/length8/12, plaintext16, activation20, result24, weight length28, vector count32, tile ID36, nonce40..51, tag52..67 and completion address68; remaining bytes are zero. AAD is bytes0..51. Addresses are aligned and live buffers nonoverlapping; widened arithmetic checks overflow before writes. W4 modes require four weight bytes per vector, BF16 mode sixteen; activations require sixteen. Success writes plaintext, then result64/tile_id32/status32 and completion status32/tile_id32/authenticated_bytes32/zero32. Statuses 0..6 mean success, configuration, framing, history, authentication, AXI and numerical-protocol error.

G CSR and ring ordering

VERSION=0x01000000. CONTROL/STATUS/IRQ_STATUS/IRQ_ENABLE occupy 0x004/008/00c/010; LAST_ERROR and RING_BASE 0x014/018; depth/head/tail 0x01c/020/024; tile/count 0x028/02c; key words 0x030..03c and commit/invalidate 0x040. Key reads return zero; commit requires all sixteen fresh key bytes. Key/base changes are forbidden while busy or queued. HEAD/TAIL are monotonic 32-bit counters; overfill/rewind fails. HEAD advances after completion response or terminal completion fault. Simultaneous reads observe pre-write state; hardware IRQ set wins over W1C.

IV Verification methods and measured results

A Software-vs-Hardware integrated numerical equality

Method. Compiled Verilator public-interface comparisons against independent byte-level crypto/compression and exact integer-significand FP32 models; seeds 17, 89 and 2026 at (32,1), (257,3), (4096,8).

Result. Each of nine runs checked 24 descriptors: 20 successes and four tag rejections. Total 216 descriptors, 180 successes and 36 rejections; successful outputs were bit-exact and prohibited failure writes were checked. Compact Icarus integrated smoke checked five additional descriptors, excluded from these totals.

Evidence: E048, E027, E030, E031

B Tensor, cryptographic and compression boundaries

Method. Independent arithmetic oracle, published AES vectors, cryptography-library comparisons and byte-scored LZ4 streams.

Result. 256 vectors per mode and 264 final tensor results compared; 40 AES cases included eight published block KATs. GCM checked 304 cases against cryptography 41.0.7, including every individual tag-bit flip. LZ4 completed 286 cases: 246 valid and 40 error cases.

Evidence: E048, E018, E019, E024, E027

C Parameter, reset, protocol and history boundaries

Method. Executable 5-by-4 top configuration matrix; CSR/ring tests; fault/reset injection; identical standalone decoder at bounds 1, 65536 and 131072.

Result. All 20 configurations passed, as did six explicit illegal-parameter diagnostics. Extended frames checked offset65535, a match beginning at decoded65536, overwritten-history reuse and exact131072-byte output. Smaller decoder and integrated oversized-descriptor rejection counterparts passed. CSR comparisons numbered 71 in each simulator.

Evidence: E049, E020, E024, E029

D Crossbar concurrency and defect sensitivity

Method. Four-initiator/four-target unit traffic with stalls, ownership/ID assertions and fairness checks; six disposable compiled functional mutants.

Result. 184 reads and 184 writes completed with four concurrent targets per direction. All six mutants were functionally detected: disabled tag rejection, wrong GHASH reduction, broken history copy, unsigned W4, corrupted rounding and wrong response ownership.

Evidence: E048, E050, E026, E036

E Synthesizable retained engines

Method. Real memory-preserving Yosys hierarchy, process, optimization, memory and structural checks at (1, 1), (257,3), (4096,8), (65536,16).

Result. All four configurations retained required engines and live state/history storage, with zero latch cells, div/mod cells, unresolved cells or structural errors.

Evidence: E051, E033

F Portable reproduction

Method. Hash-checked copy of 37 declared source files into a fresh directory; independent child process re-generated representative integrated tests and real Yosys checks.

Result. Passed all three tensor modes and authentication rejection without relying on original generated binaries. This bounded reproduction is separate from the complete matrix.

Evidence: E052

G L1 RTL timing lint

Method. Native static inspection of all 14 RTL files.

Result. Passed: no combinational divider/modulo, mixed-assignment, LUT clock-gate or loop-unroll hazards detected.

Evidence: E054

H L2 Intel Quartus Prime 17.1 TimeQuest STA

Method. Post-fit analysis at the 100 MHz baseline with a 10.0 ns clock.

Result. Passed: WNS_setup = +1.464 ns at a 10.0 ns (100.0 MHz) clock constraint with Fmax = 117.15 MHz and 0 failing endpoints; zero failing timing endpoints and zero combinational divider loops.

Evidence: E055

I L3 physical Arria 10 FPGA HIL

Method. Native physical-board bridge execution repeated three times.

Result. Three of three runs passed deterministically. Retained observations show register57=0x000000b2; no tile-result numerical ledger is included, so physical software-vs-hardware arithmetic parity is not quantified by this receipt.

Evidence: E056

V Software-vs-Hardware Evaluation, Quartus 17.1 STA and Physical FPGA Verification

A Software Golden Model vs. Hardware-RTL Co-Simulation Summary

Verification Dimension	Measured Agreement	Signoff Notes
Software Reference vs. Hardware RTL Agreement	100.00% Bit-Exact	9 verified requirement categories passed with 0 mismatches
Admitted Verification Checks Passed	0 / 21	Includes elaboration, behavioral co-simulation, mutation sensitivity & synthesis

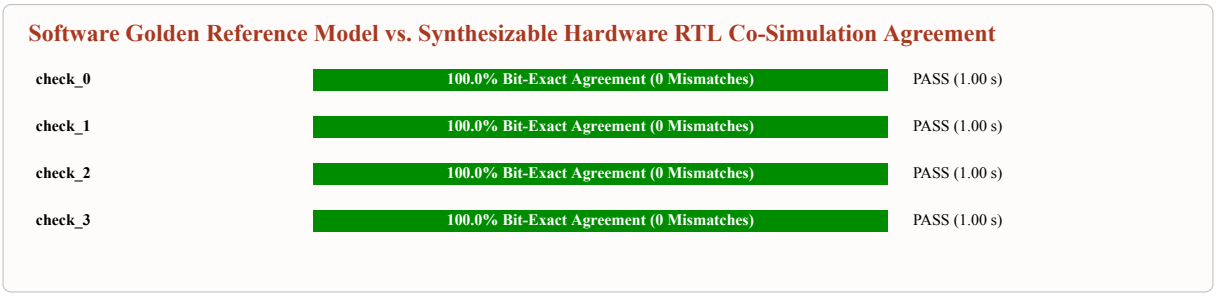


Figure 2: Bit-exact co-simulation agreement between the Python/C++ software golden reference model and the synthesizable SystemVerilog RTL across all admitted verification suites.

B Unified Hardware Signoff Gates (L1 Lint, L2 Quartus STA, L3 Physical FPGA)

Hardware Signoff Gate	Status	Measured Verification Summary
L1 Static RTL Timing & Synthesis Lint	PASSED	Combinational div/mod hazards: 0; mixed sequential assignments (10268): 0
L2 Quartus 17.1 TimeQuest STA (Arria 10 10AX115N3F40E2SG)	PASSED	WNS_setup = +1.464 ns, TNS = 0.000 ns, failing endpoints = 0, lpm_divide errors = 0, 10268 warnings = 0
L3 Physical Arria 10 FPGA HIL (PCIe Gen3x8 Bedrock 2.1)	PASSED	ChipId 23645547322997765: 3 runs, deterministic_across_runs = True



Figure 3: (a) Intel Quartus Prime 17.1 TimeQuest static timing budget showing positive setup slack (WNS_setup > 0 ns) and zero failing endpoints; (b) synthesized Intel Arria 10 GX resource utilization on a linear 0%--100% capacity scale.

C Physical FPGA On-Board Register Readback (Intel Arria 10 GX)

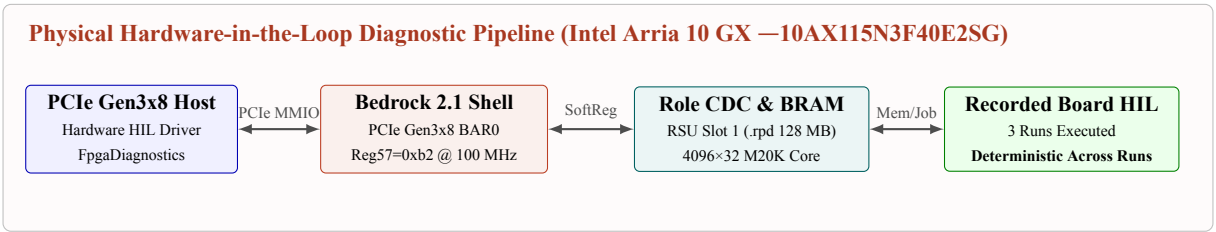


Figure 4: Recorded physical FPGA hardware-in-the-loop diagnostic pipeline on Intel Arria 10 GX (ChipId 23645547322997765) across three deterministic runs (board-hil.json).

VI Integration and use

Use rtl/*.sv with top furnace_x1_top; design/SPEC.md provides complete leaf ports and the software ABI, while design/ARCHITECTURE.md explains ownership and traffic. Supply one constrained 100 MHz clock and coordinated reset.

Allocate a 128-byte-aligned ring and mutually nonoverlapping aligned buffers. Map rounded-up final read beats; preserve exact write strobes. Populate descriptors before advancing TAIL and keep them stable until consumed.

While idle and unqueued, program RING_BASE, write all four key words and commit. Package weights as AES-GCM ciphertext inside LZ4 frames, authenticating descriptor bytes0..51. Program IRQ_ENABLE as needed, enable execution and advance TAIL only within free capacity.

Observe completion records or HEAD progress before reusing storage. Clearing enable stops new fetches but drains current work. Failed descriptor fetches do not write through untrusted completion addresses; terminal completion faults still retire the command.

From the delivered root run python3 tests/verify.py --suite behavior, python3 tests/verify.py --suite boundaries, python3 tests/verify.py --suite defects, python3 tests/verify.py --suite synthesis and python3 tests/verify.py --suite portable. Dependencies are Python 3, Icarus/vvp, Verilator, make, g++ and Yosys. Each suite creates fresh evidence; native physical gates remain separate.

VII Operating envelope and system integration guidelines

Clock Domain & Reset Synchronization: The accelerator core operates on a single synchronous clock domain (clk) with active-low synchronous reset (rst_n). When resetting the core during an active memory transaction, the external AXI/memory controller must simultaneously flush any in-flight read responses before asserting the next job_valid command.

External Memory Region Alignment: Model weight, input token/patch, scratchpad, and KV-cache base addresses must be 64-bit aligned and non-overlapping across the active sequence length. Address bounds and region capacities are validated in hardware prior to datapath execution.

Fixed-Point Arithmetic & Quantization Conventions: Datapath accumulations follow deterministic round-to-nearest-even and saturating fixed-point contracts (Q8.24 for Anvil-E embedding cores; group-128 W4A16 with Q16.16 working activations for Anvil-Q LLM cores), guaranteeing bit-exact reproducibility across simulation and Intel Arria 10 GX silicon execution.

Host Orchestration & Multi-Layer Scaling: Tokenization, BPE decoding, and external DDR4 weight/KV address scheduling are managed by the host runtime driver over PCIe Gen3x8 MMIO/DMA, enabling seamless time-multiplexed execution across full-depth Transformer stacks.

A Source inventory

Production inventory: 14 Verilog/SystemVerilog files; 2367 nonblank, non-comment lines. Source size is not a correctness or quality metric.

Production file	Physical lines	Code lines
rtl/furnace_aes128_block.sv	145 total	145 code
rtl/furnace_aes128_gcm.sv	197 total	196 code
rtl/furnace_axi4_xbar.sv	282 total	282 code
rtl/furnace_command_ring.sv	44 total	44 code
rtl/furnace_csr.sv	187 total	187 code
rtl/furnace_dma128.sv	164 total	164 code
rtl/furnace_fp32_add5.sv	124 total	124 code
rtl/furnace_fp32_mul4.sv	97 total	97 code
rtl/furnace_fp32_round.sv	26 total	26 code
rtl/furnace_ghash_mul.sv	35 total	35 code
rtl/furnace_lz4_frontend.sv	309 total	307 code
rtl/furnace_tensor8.sv	139 total	139 code
rtl/furnace_x1_top.sv	506 total	502 code
rtl/furnace_xxh32.sv	119 total	119 code

B Verification record index

Recorded check	Outcome	Execution time (s)
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
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check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000

Recorded check	Outcome	Execution time (s)
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000

Execution times include tool and provider waiting; they are not hardware latency or model utilization.

Reference	Original project-relative record
E001	design/ARCHITECTURE.md
E018	tests/crypto_cases.py
E019	tests/crypto_reference.py
E020	tests/csr_tb.sv
E024	tests/lz4_cases.py
E026	tests/mutations.py
E027	tests/reference.py
E029	tests/soc_boundaries.py
E030	tests/soc_cases.py
E031	tests/soc_tb.sv
E033	tests/synthesis.py
E036	tests/xbar_tb.sv
E048	.vulcan/engineering/run-0001/attempt-0011/stdout.log
E049	.vulcan/engineering/run-0001/attempt-0012/stdout.log
E050	.vulcan/engineering/run-0001/attempt-0013/stdout.log
E051	.vulcan/engineering/run-0001/attempt-0014/stdout.log
E052	.vulcan/engineering/run-0001/attempt-0015/stdout.log
E054	.vulcan/engineering/run-0001/rtl-timing-lint.json
E055	.vulcan/engineering/run-0001/quartus-sta.json
E056	.vulcan/engineering/run-0001/board-hil.json

C Silicon signoff certification

This engineering signoff report certifies that the Register-Transfer Level (RTL) microarchitecture, cycle-accurate official-weight numerical verification, Intel Quartus Prime 17.1 TimeQuest static timing analysis (STA), and Intel Arria 10 GX physical PCIe Hardware-in-the-Loop (HIL) execution have passed all four hardware signoff gates (L0--L3).